

## 第48届世界技能大赛网络系统管理项目精英班 集训日志（教练）

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| 日期   | 2026.04.14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 教练姓名 | 黄道金                              |
| 训练模块 | Linux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 训练任务 | Linux iSCSI client configuration |
| 训练目标 | <ul style="list-style-type: none"> <li>• 掌握在 Debian 13 系统中安装与启动 iSCSI initiator 客户端的方法。</li> <li>• 理解 discovery、node、session、login、auto startup 等 iSCSI client 关键概念及其流程。</li> <li>• 掌握使用 open-iscsi 发现 target、配置 CHAP、登录 LUN 并启用自动连接的方法。</li> <li>• 能够对登录后的新块设备完成识别、分区、格式化、挂载与常见故障分析。</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |                                  |
| 训练内容 | <ul style="list-style-type: none"> <li>• iSCSI client 基础介绍与 initiator 工作流程说明</li> <li>• open-iscsi 安装、IQN 查看与 target discovery 验证</li> <li>• CHAP 登录、session 管理、自动重连与 node 配置</li> <li>• 新块设备识别、文件系统创建、挂载与问题扩展</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |                                  |
| 过程记录 | <h3>Linux iSCSI Client Configuration</h3> <p>An iSCSI client, also called an initiator, connects to a remote target server and uses exported storage as a local block device. After a successful login, the client operating system can detect a new disk and prepare it for use.</p> <p>This workflow is different from directory-based network sharing. An initiator does not mount a ready-made shared folder first. Instead, it discovers a target, logs in to a LUN, and then creates partitions or file systems on the client side when required.</p> <p>In this task, we will learn how to configure an iSCSI initiator on Debian 13, discover a target, authenticate with CHAP, identify the new disk, and make the storage available after reboot.</p> <h4>1. Task: Install the iSCSI initiator and complete the basic deployment.</h4> <ul style="list-style-type: none"> <li>• Install the iSCSI initiator package on the Debian 13 client.</li> <li>• Check the local initiator IQN.</li> <li>• Start the required services for session handling.</li> <li>• Verify that the client is ready to perform discovery.</li> </ul> <h5>1.1. Principles</h5> <ul style="list-style-type: none"> <li>• Which package provides iSCSI initiator tools on Debian 13?</li> </ul> |      |                                  |

- What is stored in `/etc/iscsi/initiatorname.iscsi` ?
- What is the role of `iscsid` during iSCSI client operation?
- Which commands can be used to verify the current client configuration?

## 1.2. Skills

1. Install `open-iscsi` by using the `apt` package manager.
2. Check the local initiator identity in `/etc/iscsi/initiatorname.iscsi` .
3. Start or inspect `iscsid` and `open-iscsi` services with `systemctl` .
4. Use `iscsiadm -m node` and related commands to confirm that the initiator tools are available.

## 2. Task: Configure the core features of the iSCSI client.

- Discover the target from the storage server.
- Save the discovered node information.
- Log in to the specified target.
- Confirm that the login creates a usable session.

### 2.1. Principles

- What is the difference between discovery and login?
- What is a node record on the client side?
- Why does a successful session matter before looking for a new disk?
- Which commands are most useful for checking whether a LUN is attached?

### 2.2. Skills

1. Use `iscsiadm -m discovery -t sendtargets -p <server-ip>` to find the exported target.
2. Review the discovered target in the node database with `iscsiadm -m node` .
3. Log in to the target by using `iscsiadm -m node -T <target-iqn> -p <server-ip> --login` .
4. Confirm the active session with `iscsiadm -m session` before continuing to disk preparation.

## 3. Task: Configure authentication, permissions, and access behavior.

- Match the client IQN with the ACL configured on the server.
- Configure CHAP credentials for the selected node.
- Enable automatic login after reboot.
- Verify that the initiator reconnects correctly when the system starts.

### 3.1. Principles

- How do client IQN and server ACL work together?
- Which node parameters are used for CHAP authentication?
- Why should automatic startup be configured for production-style use?
- Which mistakes commonly prevent reconnect after reboot?

### 3.2. Skills

1. Update node parameters such as `node.session.auth.authmethod`, `node.session.auth.username`, and `node.session.auth.password`.
2. Set `node.startup` to `automatic` for the desired target node.
3. Test logout and login again to confirm that the saved node settings work correctly.
4. Verify that incorrect CHAP credentials cause login failure and that corrected values restore access.

## 4. Task: Restrict access, validate security behavior, and complete an advanced scenario.

- Identify the new disk after login.
- Create a partition or file system on the exported LUN.
- Mount the storage safely and persist it across reboots.
- Analyze failures related to device discovery, authentication, or boot-time mounting.

### 4.1. Principles

- Which commands help distinguish the new iSCSI disk from an existing system disk?
- Why should UUID-based mounting be preferred for persistent storage use?
- What is the purpose of `_netdev` in `/etc/fstab` for network-backed block devices?
- Which checks should be performed when the target login succeeds but no expected mount is available?

### 4.2. Skills

1. Use `lsblk`, `blkid`, `dmesg`, and `/dev/disk/by-path/` to identify the new iSCSI device correctly.
2. Create a file system such as `ext4` on the exported partition or raw LUN when appropriate.
3. Mount the storage to a directory such as `/data` and validate access with `df -h`.
4. Add a persistent entry to `/etc/fstab` with options such as `_netdev` and verify it with `mount -a`.

## 5. Extensions

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|     | <ol style="list-style-type: none"><li>1. How can an iSCSI disk be turned into an LVM physical volume on a client?</li><li>2. What are the operational differences between mounting a raw LUN and using a partition table?</li><li>3. How can <code>multipath-tools</code> improve resilience in a storage network?</li><li>4. Why should the same ordinary file system on one LUN not be mounted read-write by multiple hosts without cluster coordination?</li><li>5. How can iSCSI storage be prepared for database or virtualization workloads?</li></ol> <h2>6. References</h2> <ul style="list-style-type: none"><li>• <a href="#">Wikipedia: iSCSI</a></li><li>• <a href="#">Debian Wiki: Open-iSCSI</a></li><li>• <a href="#">Debian Manpage: iscsiadm</a></li></ul> |
| 总结  | <p><b>训练目标达成情况：</b></p> <p>从本次训练完成情况看，选手基本能够独立完成 Debian 13 环境下 iSCSI client 的安装配置，并根据任务要求完成 target 发现、认证登录、磁盘识别和基础挂载验证，说明在网络块设备接入与使用方面已具备初步实操能力。</p> <p><b>不足之处：</b></p> <ul style="list-style-type: none"><li>• 部分选手对 discovery、login、session 与本地块设备出现之间的关系理解还不够清晰，容易把发现成功误认为登录成功。</li><li>• 个别选手能够完成基本连接，但对 node 参数持久化、自动登录和 `_netdev` 挂载细节掌握还不够稳定。</li></ul> <p><b>改进方向：</b></p> <ul style="list-style-type: none"><li>• 下一阶段应加强针对认证错误、设备名判断错误和开机自动连接失败等场景的专项排查训练。</li><li>• 增加 iSCSI 磁盘结合 LVM、文件系统管理和多路径接入的综合实践内容。</li></ul>                                                                                                                                                                                                                                            |
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